

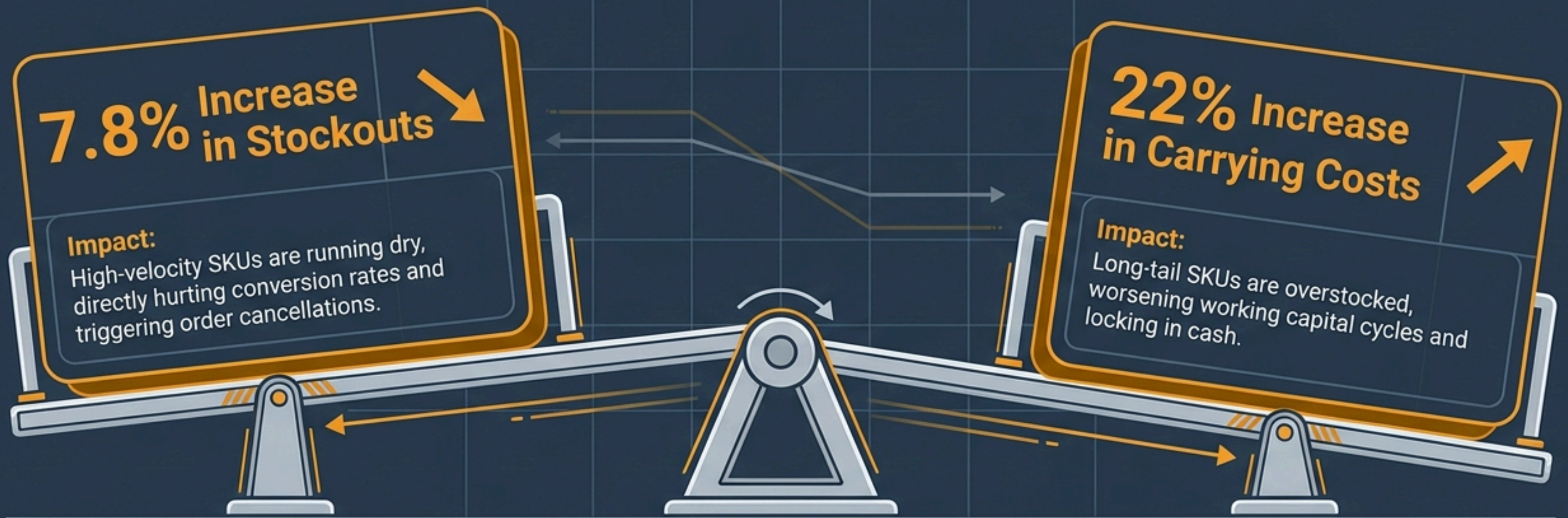
Predictive Inventory Optimisation for E-Commerce Fulfilment

A Data-Driven Approach to Demand Forecasting and Warehouse Replenishment

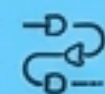


Widening Inefficiencies at the East Zone Fulfilment Centre

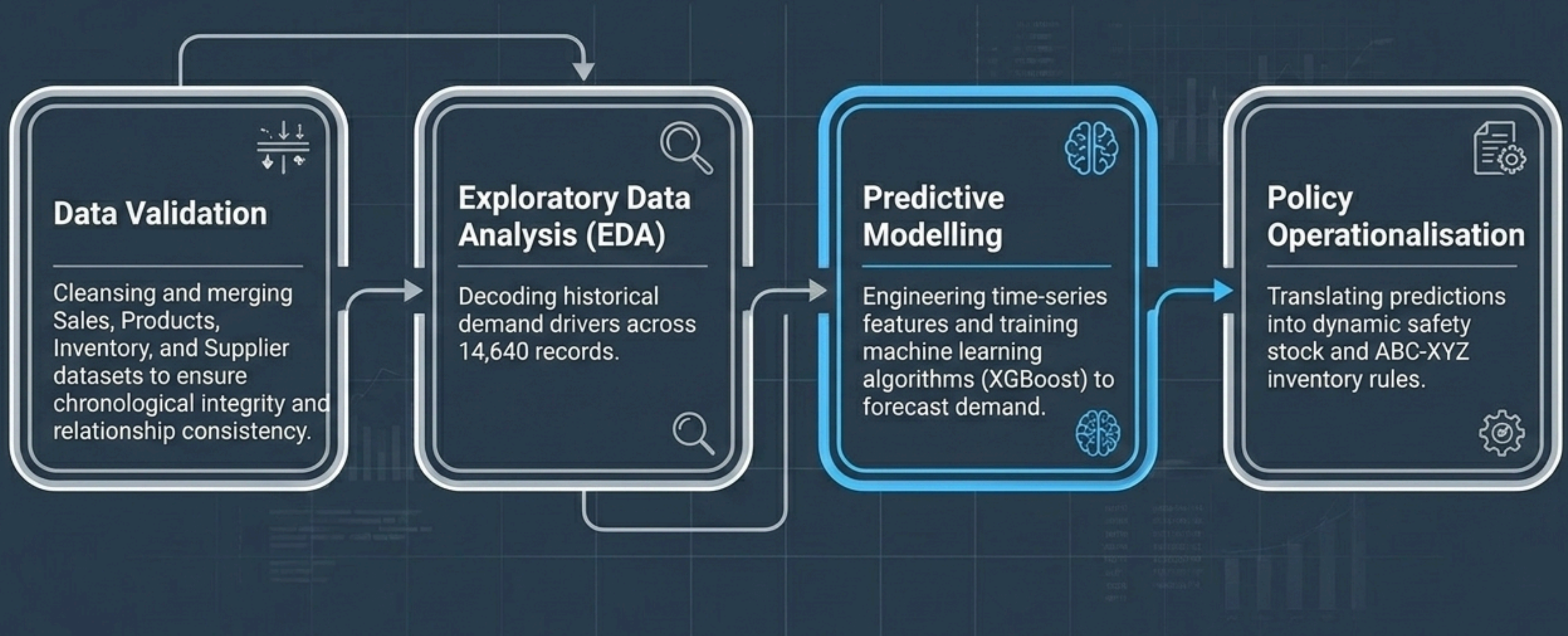
NovaCart's EZFC serves a portion of 12 million monthly shoppers. Promised SLAs (Next-Day Delivery, Same-Day Dispatch) are failing due to an imbalanced inventory system.



The Mandate: Shift from reactive, manual replenishment to a predictive, automated inventory system.



Translating Raw Data into Automated Warehouse Actions



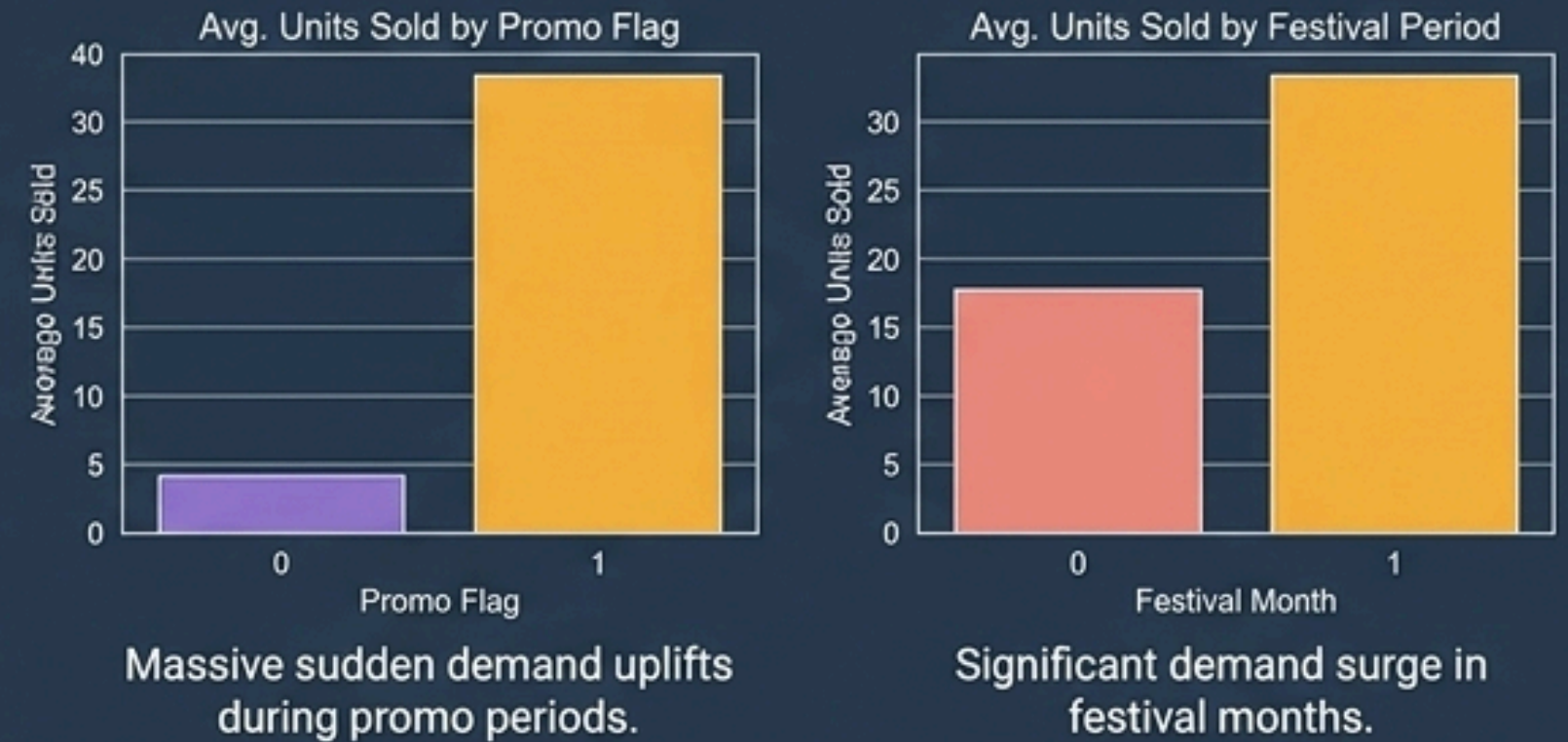
Identifying the Hidden Triggers of Demand Volatility

Category Volume Drivers



Electronics Accessories and Home & Kitchen dominate the volume, showing significant concentration.

External Triggers

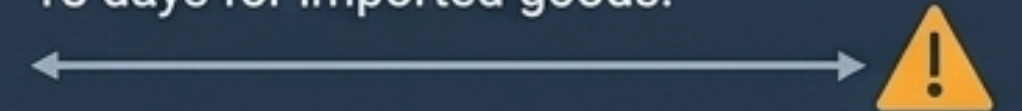


Supplier Dynamics

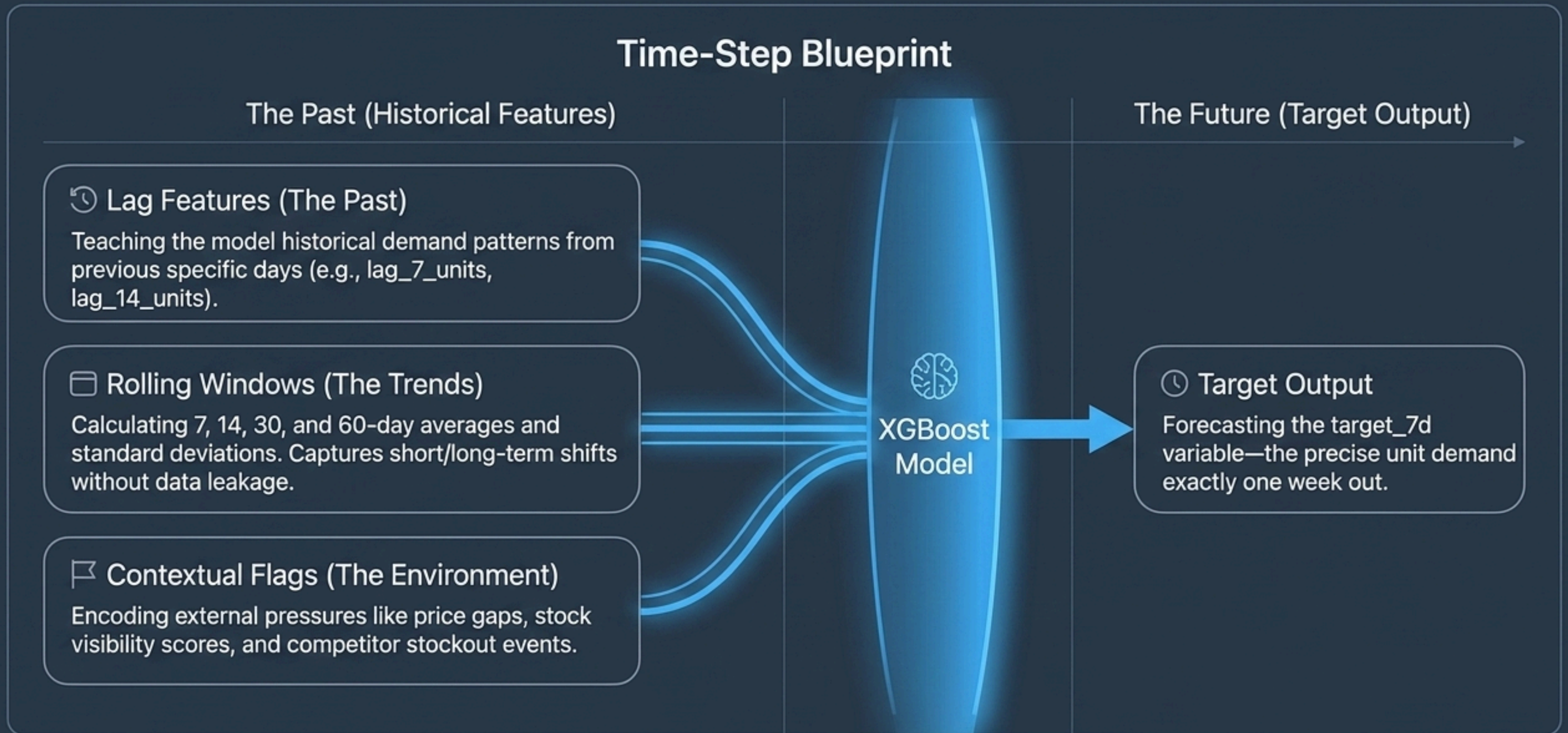


Extreme Supplier Lead Time Variance

Ranges from 2 days for local vendors to 18 days for imported goods.



Engineering Predictive Signals from Historical Patterns



Isolating the Optimal Demand Forecasting Engine

Baseline (Naive Method)

Assumes demand matches the previous week.

Result: R^2 -0.1924

Completely fails to capture sudden promotional spikes or seasonal shifts.

Random Forest

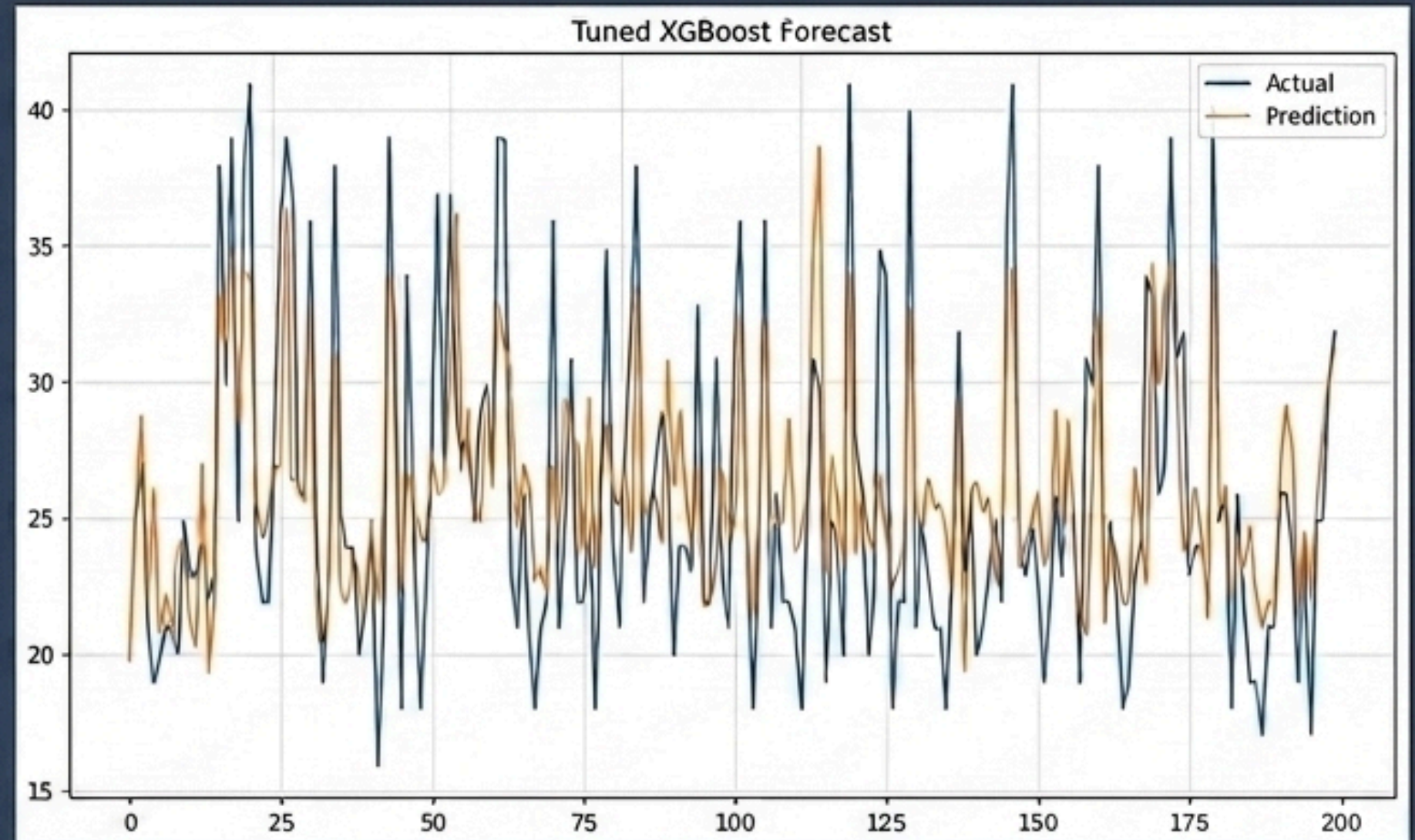
Captures non-linear relationships using multiple decision trees.

Result: Stronger performance, but lacks ultimate precision in extreme variance scenarios.

Tuned XGBoost (The Winner)

Sequential tree-building optimized via hyperparameter tuning.

Result: R^2 lifted to 0.5922. Lowered **MAE** and **RMSE**, yielding the highest generalisation capability.
RMSE: 5.01, MAE: 3.89



Strategic Inventory Segmentation via ABC-XYZ Classification

Categorising SKUs to dictate replenishment strategy.

ABC Analysis (Value)



Based on Annual Consumption Value (Demand $\times$ Price). Just 23 'A' class SKUs drive the bulk of network revenue.

XYZ Analysis (Volatility)

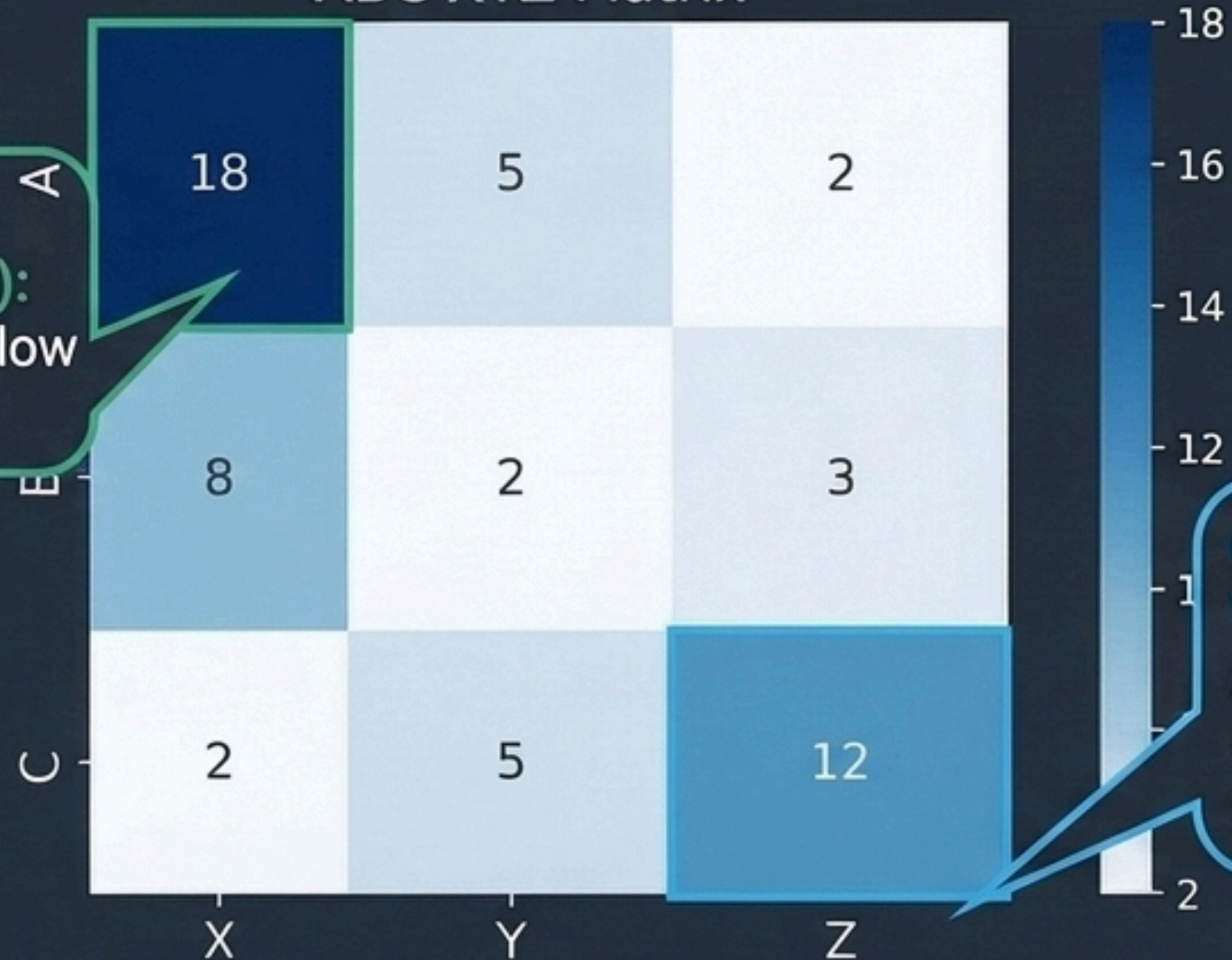


Based on the Coefficient of Variation (Demand Std / Average Demand). Separates steady demand from highly volatile items.

Insight:

High-value, highly volatile SKUs require the most dynamic, heavily monitored safety stock interventions.

ABC-XYZ Matrix



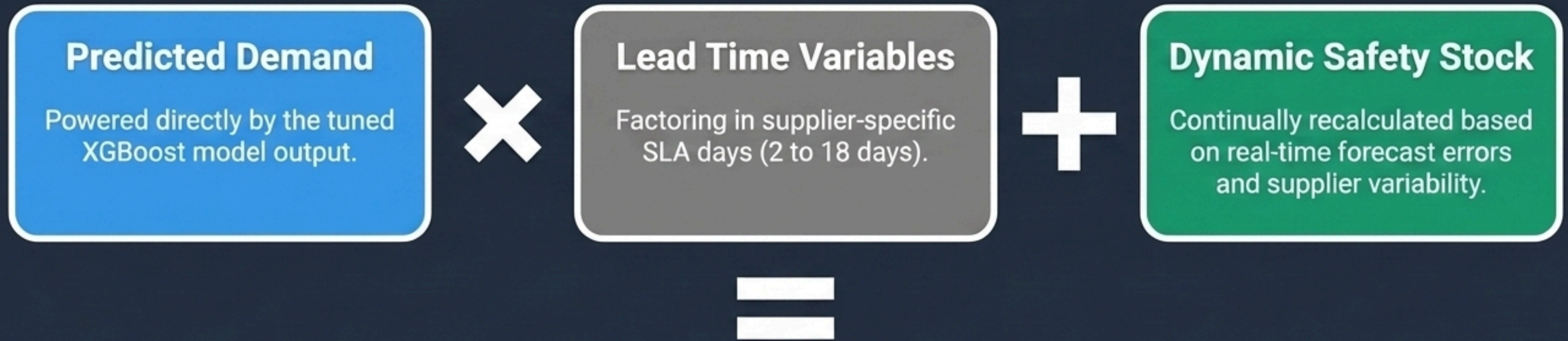
A-X (High Value, Stable):
Tight control, low safety stock.

C-Z (Low Value, Volatile):
Automated ordering, high safety buffer.

Conclusion: What We Get

A robust framework for tailored inventory management, allowing for prioritized focus on high-value, high-volatility items while automating and simplifying the approach for low-risk SKUs. This leads to optimized resource allocation and improved service levels.

Activating the Dynamic Inventory Replenishment Policy



Reorder Point (ROP) = (Avg Predicted Demand × Lead Time Days) + Calculated Safety Stock

Real-Time Example from Notebook (SKU X): (120 units/day × 14 days) + 350 units = 2030 units ROP

Takeaway: Static thresholds are retired. The system now adapts daily to both market demand and supplier performance.

Quantifying and Managing SKU-Level Inventory Risk

$$\text{Risk Score} = (\text{Demand Standard Deviation}) \times (\text{Lead Time Variability})$$

High Risk



14 SKUs

High demand volatility combined with unreliable supplier delivery times.

Medium Risk



13 SKUs

Moderate fluctuations.

Low Risk

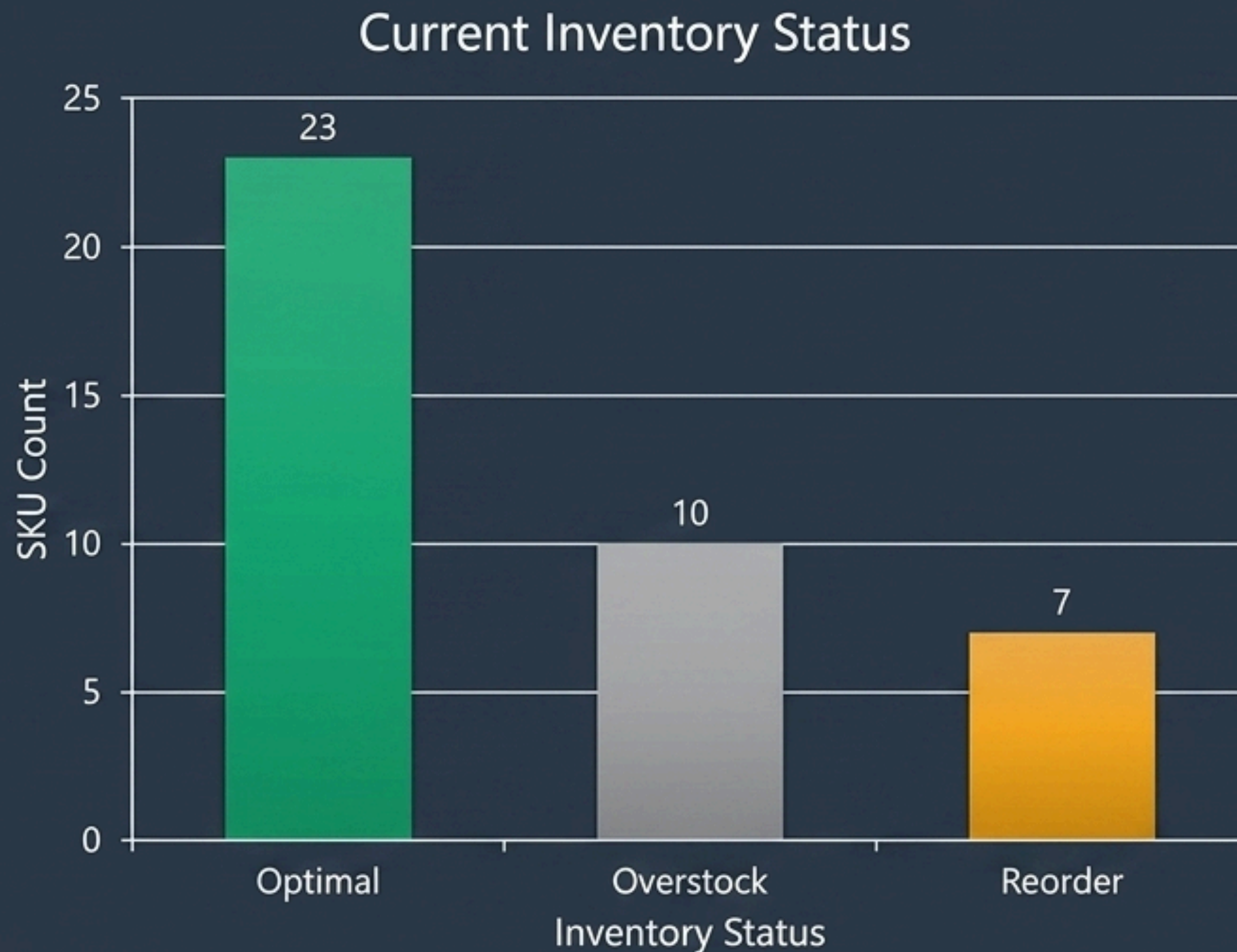


13 SKUs

Highly predictable demand and stable local suppliers.

Operational Directive: High-risk SKUs automatically trigger buffer stock prioritisation to mathematically protect the $\geq 96\%$ service level target.

Applying the New Policy Uncovers Immediate Action Areas



✅ Optimal (~23 SKUs)

Perfectly balanced according to the new dynamic thresholds.

↓ Overstock (~10 SKUs)

Currently tying up unnecessary working capital. Recommended for immediate promotional clearance or delayed reordering.

⚠️ Reorder (~7 SKUs)

Sitting below the new dynamic safety thresholds. At immediate risk of stockout and require urgent purchase orders.

Strategic Outcomes and Operational Recommendations



Service Levels Protected

By deploying the **Tuned XGBoost forecasting** and dynamic safety stock, the network directly targets the reduction of the current **7.8%** stockout rate, protecting the **≥96% SLA**.



Working Capital Freed

Systematic identification of **'Overstock'** SKUs empowers the Finance team to achieve the mandate of reducing **excess inventory by ≥15%**.



Operations Ready

The **predictive pipeline** is fully adaptable. It seamlessly manages varying cadences, from 2-day local vendors to 18-day imports, feeding directly into NovaCart's **daily replenishment workflow**.